

Time Dilation

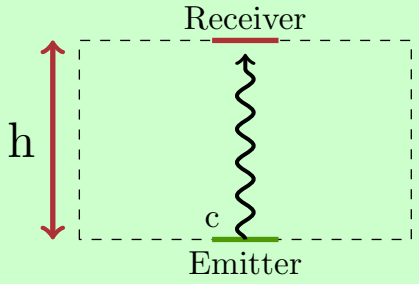
Review: Last time we started talking about Special Relativity

1. What is special about special relativity?
2. What are the postulates of SR?
3. Do all observers agree on the order of two events?
4. What is a simultaneous measurement?
5. What are proper distance and proper time?

Worksheet time

Imagine a laser is emitted from the bottom of a train and detected at the top. The train has velocity v.

(a) Aboard Train

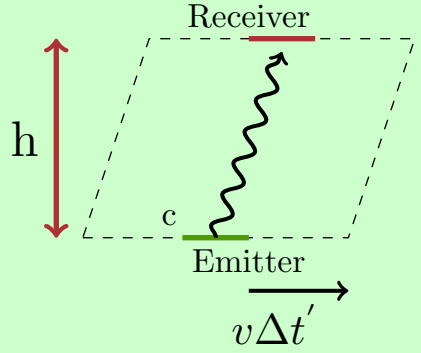


τ = time to receiver

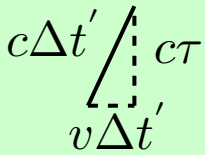
$$h = c\tau$$

$$\begin{aligned}
\left(c\Delta t'\right)^2 &= (c\tau)^2 + \left(v\Delta t'\right)^2 \\
\left(c\Delta t'\right)^2 - \left(v\Delta t'\right)^2 &= (c\tau)^2 \\
c^2\Delta t'^2 - v^2\Delta t'^2 &= c^2\tau^2 \\
\left(c^2 - v^2\right)\Delta t'^2 &= c^2\tau^2 \\
\Delta t'^2 &= \left(\frac{c^2}{c^2 - v^2}\right)\tau^2 \\
\Delta t'^2 &= \left(\frac{1}{1 - (v/c)^2}\right)\tau^2 \\
\Delta t' &= \left(\frac{1}{\sqrt{1 - (v/c)^2}}\right)\tau
\end{aligned}$$

(b) Outside Train



t' = time to receiver



Define $\gamma = \dfrac{1}{\sqrt{1 - \beta^2}}$

For any frame uses this formula
But you need to figure out which frame has the proper time

$$\Delta t = \gamma \Delta \tau$$

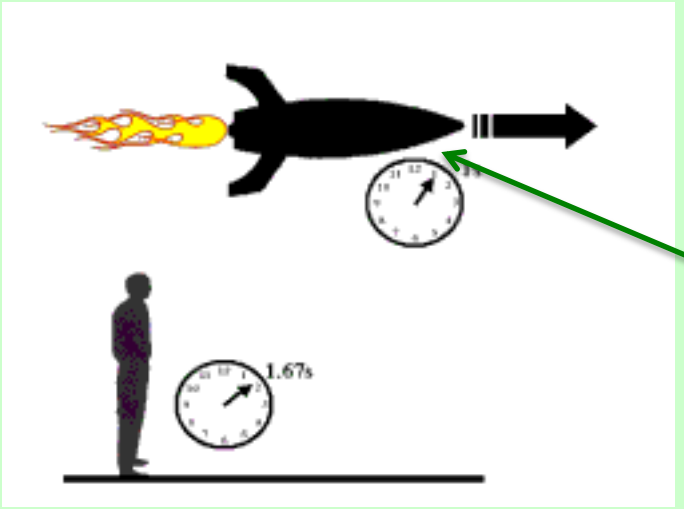
Time Dilation

Consequence #3: Time Dilation

Time Dilation : **Moving clocks run slow**

A) Clocks on the spaceship run slow according to the ground.

B) But run at the normal rate aboard the ship



I see as Slower

Ex. If the spaceship is moving at 0.8c, how much time elapsed aboard the spaceship?

Who has the proper time?

1. Question is about time on spaceship the man watches him from afar
2. Usually the proper time is the one that makes a journey (but not always)
3. Proper time is the time recorded by the same clock at different times.

Time Dilation Equation

$$\Delta t = \gamma \Delta \tau$$

Time dilated time ("Outside" time)

Proper Time ("Inside" time)

Ex. Saturn is 1.43×10^{12} m from the sun. A rocket travels along a line from the sun to Saturn at a constant speed of 0.9c relative to the solar system.

A. Draw a pictorial representation of the journey according to an observer on Earth

B. Draw a pictorial representation of the journey according to an astronaut on the ship

C. How long does the journey take according to the observer on the Earth?

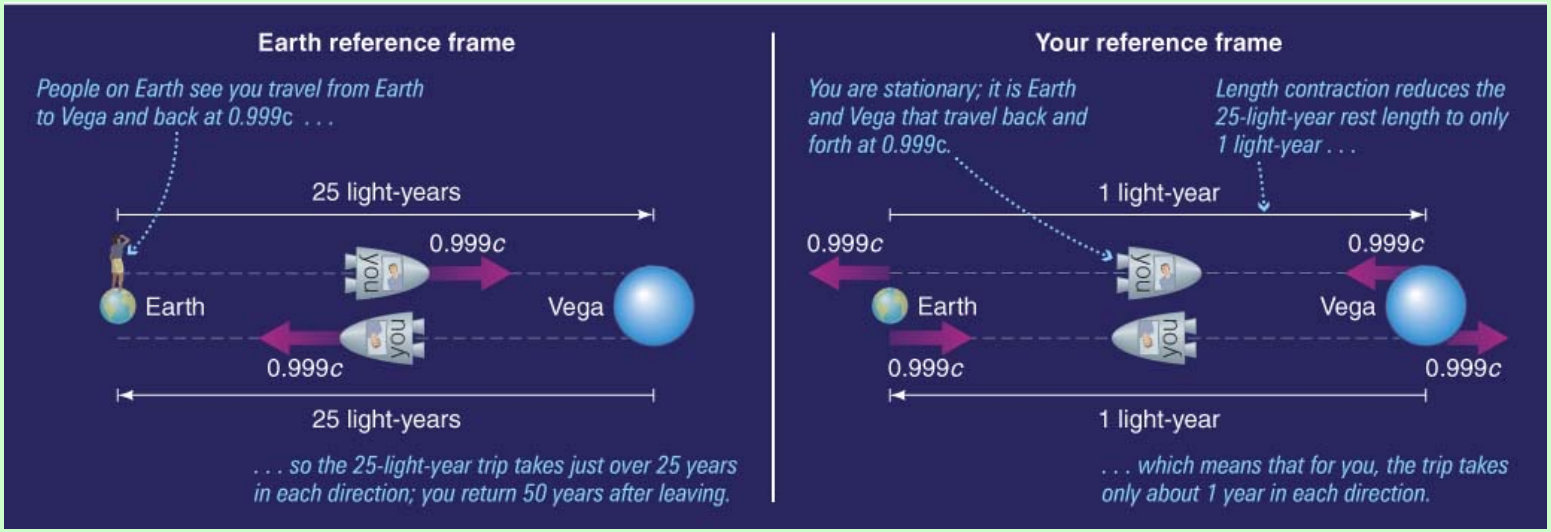
D. How long does the journey take according to the Astronaut on the ship?

Consequence #4: Length Contraction

Length Contraction: A meter stick moving relative to you is shorter than your meter stick in its direction of motion

Length Contraction Equation

$$L = \frac{L_0}{\gamma}$$



Distance seems to be shortened according to the spaceship

- Your clock ticks slow, so you measure less distance has passed
- **Who has the proper distance L_0 ?** Need to ask where distance is actually measured

Ex. Saturn is 1.43×10^{12} m from the sun. A rocket travels along a line from the sun to Saturn at a constant speed of 0.9c relative to the solar system.

A. What is the distance between the Sun and Saturn in the solar system frame?

B. What is the distance between the Sun and Saturn in the ship's frame?